

# HYPERBOLIC GEOMETRY

JOSEPHINE XIE & TOYESH JAYASWAL

## 1. SOME TOPOLOGY AND GEOMETRY

It is a famous fact that a torus can be obtained by identifying opposite sides of a square. First, you identify two sides and get a cylinder with the other two sides becoming circles which need to be identified. Identifying these two circles gives us a torus.

We can interpret the identification of opposite sides of a square as follows: we tile the plane with squares which are all translates of each other. The square is identified a fundamental domain of the plane i.e. a region where there is a unique representative for each point of translations (similar to finding a set of representatives for  $\mathbb{Z}[i]/a$ ), but there is redundancy on the edges, so we need to identify them.

Now, the plane is flat, and this actually allows us to construct a "flat torus". In the first step when we fold the square into the cylinder, the cylinder is still "flat" in some sense that although it curves in one direction, it does not curve in the other and we were able to get from the square to the cylinder without stretching it. However, to get from the cylinder to the torus, we need to stretch it so that the outside circle is larger than the smaller circle.

However, we can modify our construction to get a genuinely flat torus: we can flatten our cylinder to get a "double covered rectangle". Then, we can fold the rectangle in half and glue the circles (which are now "double edges") to get a "quadruple covered square". You can still imagine what it would mean to rotate this "torus" in each direction - the sheets slide on top of each other - and so we do still have a genuine torus, but now it's actually flat! If you think this is cheating, as a little note, we can identify the torus as  $S^1 \times S^1$ , the product of two circles, and we can thus embed a torus into  $\mathbb{R}^4$  as  $(\cos(\theta), \sin(\theta), \cos(\phi), \sin(\phi))$ . It turns out (using some heavy machinery beyond this talk) that this is a genuinely flat!

It's an interesting question to ask whether this can be done for other surfaces. The problem of using a plane to represent a sphere had plagued map makers for centuries as they could not seem to map the earth onto a square without stretching it. It turns out this is impossible, and there is a topological obstruction (to be discussed later) coming from the fact that a sphere is "positively curved".

But we can ask the question about general surfaces. Can we map them honestly? It turns out for  $g \geq 1$ , we can represent a genus  $g$  surface as the quotient of a  $4g$ -gon by identifying sides cleverly. To see why this is true when  $g = 2$ , we can cut the octagon along the diagonal and we get two pentagons with 4 sides identified. We can think of this as a square with a corner cut off, and when we identify the sides, we get a torus with a circle cut out. When we glue the two circles, we get a genus 2 surface.

It turns out that we can't tile the plane by octagons! This should seem non-obvious - it does not have to be a regular octagon, but an arbitrary one. It turns out that curvature again plays a role and obstructs the possibility of a tiling. But we'll see at the end that there is a salvage!

## 2. WHAT IS HYPERBOLIC GEOMETRY?

For over two thousand years, people had believed that only one geometry was possible, which was Euclidean Geometry, and they had accepted the idea that this geometry described reality.

This system consisted of a collection of undefined terms like point and line, and five axioms from which all other properties could be deduced by a formal process of logic. Four of the axioms were so self-evident that it would be unthinkable to call any system a geometry unless it satisfied them:

- A straight line may be drawn between any two points.

- Any terminated straight line may be extended indefinitely.
- A circle may be drawn with any given point as center and any given radius.
- All right angles are equal.
- If a straight line falls on two straight lines in such a manner that the interior angles on the same side are together less than two right angles, then the straight lines, if produced indefinitely, meet on that side on which are the angles less than the two right angles.

The first four postulates are easy to understand; the fifth is more complicated. It is equivalent to the following, which is now known as the parallel postulate: Given any infinite straight line and a point not on that line, there exists a unique infinite straight line through that point and parallel to the given line.

However, in the first half of the 19th century, Gauss proved that this was impossible: the parallel postulate was independent of the other four postulates. He did this by making the discovery that there exist consistent geometries for which the first four postulates hold, but the parallel postulate fails. In 1824, Gauss wrote ‘The assumption that the sum of the three sides (of a triangle) is smaller than 180 degrees leads to a geometry which is quite different from our (Euclidean) geometry, but which is in itself completely consistent.’ (One can show that the parallel postulate holds if and only if the angle sum of a triangle is always equal to 180 degrees.) This was the first example of a non-Euclidean geometry.

Gauss never published his results on non-Euclidean geometry. However, it was soon rediscovered independently by Lobachevsky in 1829 and by Bolyai in 1832. Today, the non-Euclidean geometry of Gauss, Lobachevsky and Bolyai is called hyperbolic geometry and any geometry which is not Euclidean is called non-Euclidean geometry.

### 3. CURVATURE

**3.1. What is a curve?** The words ‘function’ and ‘map’ are often used somewhat interchangeably to refer to some form of correspondence between sets  $X \rightarrow Y$  in which each element of  $X$  is sent to exactly one element of  $Y$ . In the fields of geometry and topology, the word ‘map’ has a clear distinction from ‘function’.

**Remark 3.1.** A **map** is a continuous function.

**Definition 3.2 —** A **curve** in  $\mathbb{R}^n$  is a map  $\gamma : (a, b) \rightarrow \mathbb{R}^n$  where  $-\infty \leq a < b \leq \infty$ . The **trace** of a curve  $\gamma$  is the image  $\{\gamma(t) | t \in (a, b)\}$ .

The set of points idea of a curve is exactly the trace of a curve defined as a function. Suppose that  $\mathcal{C} \subset \mathbb{R}^n$  is a set of points, and there is a curve  $\gamma$  whose trace is  $\mathcal{C}$ , then we say that  $\gamma$  parameterised the set  $\mathcal{C}$ .

#### Example 3.1

We can parameterise the parabola given by  $y = x^2$  by

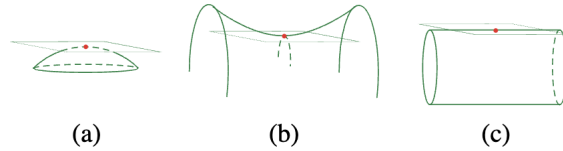
$$\gamma : (-\infty, \infty) \rightarrow \mathbb{R}^2 : t \mapsto (t, t^2)$$

**3.2. How much does a curve curve?** We want to understand how much a curve curves or bends, in other words to quantify the fact that a line doesn’t curve, a large circle (eg. the equator of the Earth) curves but very slowly, and a small circle (eg. a horse on a merry-go-round) curves quickly. Curvature in this intuitive sense is really a property of the trace of a curve, so we should define it so that the curvature does not depend on the choice of parametrisation.

**Definition 3.3 —** Let  $\gamma$  be a unit speed curve with parameter  $s$ , then its **curvature** at the point  $\gamma(s)$  is given by  $\kappa(s) = \|\ddot{\gamma}(s)\|$ .

There are three fundamental types of curvature: (a) Positive curvature: at a point if the surface lies entirely on one side of the tangent plane (b) Negative curvature: at a point if it is saddle-shaped, in the sense

that the tangent plane cuts through the surface (c) Zero curvature: the surface has a line along which the surface agrees with the tangent plane



Things you need to know for the following note:

- Curvature of a circle:  $\kappa = \frac{1}{r}$
- Curvature of a sphere:  $\kappa = \frac{1}{r^2}$
- Curvature of a hyperbolic plane:  $\kappa = -1$

Prove it if you are interested!

#### 4. PSEUDOSPHERE

**4.1. Geodesics.** Geodesics are the curves in a surface that make turns just to stay on the surface and never move sideways. A bug living in the surface and following such a curve would perceive it to be straight. A geodesic is a generalisation of the notion of a “straight line” from a plane to a surface, on which it represents in some sense the shortest path between two points.

**4.2. What is a pseudosphere?** In geometry, a pseudosphere is a surface with constant negative curvature.



Why does it have a negative curvature? If you pick any point on the surface above except for the little rim where the two halves meet, take the normal to that point and draw two orthogonal planes through that normal, you will see that one curves down on both sides and one curves upward on both sides and this is why it is considered to have negative curvature. In fact, the pseudosphere has a constant negative curvature of  $\kappa = -\frac{1}{R^2}$

#### 4.3. Conformal map of a pseudosphere. <sup>1</sup>

A conformal map is a function that locally preserves angles, but not necessarily lengths.

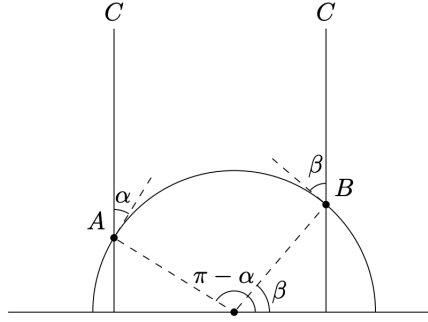
For example, a map is not accurate in the area a country but accurate in compass direction.

For the conformal map of the pseudosphere, we can tell that as the  $y$  value gets smaller, the size of the disc gets smaller. This is because the projection is from the top, and in order to preserve the ‘direction’ of the discs, the size of the disc would look smaller and smaller, but in reality, they are all same sized.

---

<sup>1</sup>Tristan Needham, *Visual Differential Geometry and Forms: A Mathematical Drama in Five Acts*, Princeton University Press, Princeton, NJ, 2021.

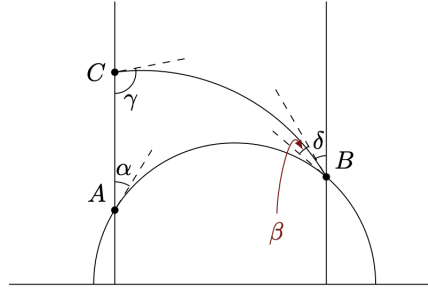




Now we have

$$\begin{aligned}
 \mathcal{A}(ABC) &= \int_{\cos(\pi-\alpha)}^{\cos\beta} \int_{\sqrt{1-x^2}}^{\infty} \frac{1}{y^2} dy dx \\
 &= \int_{\cos(\pi-\alpha)}^{\cos\beta} \frac{1}{\sqrt{1-x^2}} dx \\
 &= \left[ -\cos^{-1}(x) \right]_{\cos(\pi-\alpha)}^{\cos\beta} \\
 &= \pi - \alpha - \beta
 \end{aligned}$$

When  $C$  is not at infinity, and we can arrange  $AC$  in a vertical half-line. Also, we can move  $AB$  to  $x^2 + y^2 = 1$ , noting that this transformation keeps  $AC$  vertical.



We consider  $\triangle_1 = AB\infty$  and  $\triangle_2 = CB\infty$ . Then we can immediately write

$$\mathcal{A}(\triangle_1) = \pi - \alpha - (\beta + \delta)$$

$$\mathcal{A}(\triangle_2) = \pi - \delta - (\pi - \gamma) = \gamma - \delta$$

$$\begin{aligned}
 \Rightarrow \mathcal{A}(ABC) &= \mathcal{A}(\triangle_1) - \mathcal{A}(\triangle_2) \\
 &= \pi - \alpha - (\beta + \delta) - (\gamma - \delta) \\
 &= \pi - \alpha - \beta - \gamma
 \end{aligned}$$

Therefore, for each hyperbolic triangle  $\triangle$ , say,  $ABC$ , with angles  $\alpha, \beta, \gamma \geq 0$  (note that zero angle is possible), we have

$$\mathcal{A}(\triangle) = \pi - (\alpha + \beta + \gamma)$$

## 6. GAUSS-BONNET THEOREM

We can associate a numerical value to curvature as follows: if the surface is dome shaped or bowl shaped like a sphere, it has positive curvature, while if it looks like a saddle/pringles chip, it has negative curvature. If a surface is flat in at least one direction (like a cylinder), it has 0 curvature. The curvature can be thought of as multiplying the curvature of curves in the extremal directions, which happen to be perpendicular. For example, for a sphere, both curves "curve the same way" so either it's  $++ = +$  or  $-- = -$ , while for a saddle, we have the curves "curve in opposite directions" so  $+* = -$  and for a cylinder  $0* = 0$  or  $0* = 0$  or  $0*0 = 0$ .

We can actually assign a numerical value  $\kappa(p)$  to the curvature at a point  $p$  based on how sharply it curves. A tip of a dull pencil would have a very large curvature since it curves very quickly, while the surface of the Earth would have a very small curvature since it appears to be very flat. With this, there is a very powerful theorem (whose proof is beyond these notes) that relates the geometry of a surface to its topology called the Gauss-Bonnet formula:

$$\int_S \kappa(p) dA = 2\pi\chi(S) = 2\pi(2 - 2g).$$

In other words, the total curvature of the surface  $S$  is  $2\pi$  times the Euler characteristic, which is two minus twice the genus.

**Example 6.1**

A sphere of radius  $r$  turns out to have constant curvature of  $\kappa = \frac{1}{r^2}$  everywhere and thus  $\int_S \kappa(p) dA = \frac{1}{r^2} (4\pi r^2) = 2\pi(2 - 2 \cdot 0)$ .

**Example 6.2**

We demonstrated how to construct a flat torus. This has constant curvature 0 everywhere, so we have  $\int_S \kappa(p) dA = \int 0 = 0 = 2\pi(2 - 2 \times 1)$

**Example 6.3**

For a donut-shaped torus, the uppermost and lowermost circles have 0 curvature since they have one direction that doesn't curve. The inside has negative curvature and the outside has positive curvature. Even though the area on the outside is larger, the torus bends more on the inside and thus has more curvature. These effects exactly cancel out so that the total curvature is 0.

This explains why map makers failed for so long to make an accurate flat chart of the plane - if they could, we could make a flat sphere much like we made a flat torus, but this would violate the theorem! Similarly, any surface of genus  $g \geq 2$  has negative Euler characteristic, and thus negative total curvature. Thus, we also cannot tile the plane with  $4g$ -gons, otherwise we could make a flat genus  $g$  surface. However, we are starting to uncover a space that has constant negative curvature. There is hope that we can use this hyperbolic space to chart our surfaces. We first investigate some more of the geometry of this mysterious space, and then return to this question.

## 7. REFLECTION AND TRANSLATION ALONG LINES

We've seen that semicircles perpendicular to the horizontal axis in the upper half plane correspond to geodesics (lines) in hyperbolic space. To get a better feel for why the lines are actually lines, let's try and introduce some dynamics and see how we can reflect across these lines and translate along them.

First, let's discuss reflections. As we've seen, higher up on the upper half plane, things appear much more stretched out than they actually are, while down below, they're much more compressed. A reflection along a line (semicircle)  $L$  would have to swap the inside (below) and outside (above) of the semicircle. It turns out that reflections correspond to the very elegant geometric construction of circle inversion! Circle inversion works as follows: take a circle  $C$  of radius  $r$  and center  $O$ . Then, for a point  $x$  which is a (Euclidean) distance

$d$  away from  $O$ , send  $x$  to the point  $x'$  that lies on the same ray from  $O$  to  $x$  but is a distance  $\frac{d^2}{r}$  away from  $O$ . This is called inversion along  $C$ .

A few properties about this construction:

- Inversion along  $C$  preserves  $C$  and no other points
- Points inside  $C$  are sent outside  $C$  and vice versa
- Points near  $O$  which are close together become very spread apart outside  $C$  and vice versa
- Performing this operation twice is the identity
- If  $C'$  is another circle not passing through  $O$ , the image of  $C'$  under inversion is another circle
- If  $C'$  passes through  $O$ , its inversion becomes a line, and conversely, a line not passing through  $p$  becomes a circle passing through  $O$
- We can consider angles between intersecting lines and circles, and inversion preserves these angles

The last three points are nontrivial, but are stated without proof. Assuming point 5, point 6 can be interpreted as saying a line is a "circle through  $\infty$ " and inversion swaps  $O$  and  $\infty$ .

If we invert along a hyperbolic line, items 5, 6, and 7 say that hyperbolic lines get sent to hyperbolic lines as the horizontal line is preserved, and orthogonality with the semicircles is too. These properties make it seem quite promising that circle inversion acts a lot like a reflection. It turns out that if we take the circle inversion through a hyperbolic line, it also preserves hyperbolic distance. If the hyperbolic line is a vertical line, the reflection is just the usual reflection, which can be interpreted as inversion through a circle with infinite radius and center  $\infty$ .

There's a nice corollary to this: take a hyperbolic line  $L$  and a point  $p$ . There is a unique line  $L'$  which passes through  $p$  and is orthogonal to  $L$ . To see this, the reflection through  $L$  of a line orthogonal  $L'$  to  $L$  must remain orthogonal to  $L$ . Looking near the point of intersection of the two lines, we see that the reflection of  $L'$  must also be  $L'$ . We can thus determine the line through  $p$  perpendicular to  $L$  as follows: it must be preserved under reflection (circle inversion) through the line  $L$  and so must contain the point  $p'$  which is the reflection of  $p$  through  $L$ . Then I claim there is a unique hyperbolic line through  $p$  and  $p'$ . If they have the same  $x$  coordinate, the line is simply the vertical line passing through both of them. Otherwise, we want a circle that passes through both of them and has center on the  $x$  axis. Note that the center of any circle passing through  $p$  and  $p'$  will lie on the perpendicular bisector of the segment connecting them. This bisector intersects the  $x$  axis at one point, and this is the center of the unique line that we want.

Now, we can discuss translations in hyperbolic geometry. The way we can translate along a hyperbolic line  $L$  uses the following idea: in Euclidean geometry we can foliate (divide) the plane into lines perpendicular to a given line  $L$ , and translation along  $L$  will send these lines to each other. Similarly, we can foliate the upper half plane into lines perpendicular to  $L$  - we just showed every point must lie on a unique such line! Suppose translating a distance of  $d$  along  $L$  takes a point  $p$  on  $L$  to a point  $p'$  on  $L$ . Then, we can take the line perpendicular to  $L$  through  $p$  to the line perpendicular to  $L$  through  $p'$ . Note that if we trace the points rather than the lines, we would get very peculiar curves as we translate: since translation should fix distance and distance is compressed closer to the  $x$  axis, translating "clockwise" would make all points eventually approach the right endpoint, while translating "counterclockwise" would make all points eventually approach the left endpoint. The imaginative of you may see that the path traced out by a point  $p$  could be a circle connecting the end points of  $L$  (but isn't a geodesic since it isn't perpendicular to the  $x$  axis unless  $p \in L$ ).

Let's prove this. The key idea is that any translation can be composed of two different reflections. In Euclidean geometry, if we take two parallel lines a distance of  $a$  apart and reflect across them in succession, this results in the transformation of translation along the perpendicular line of a distance  $2a$ . Now, take two hyperbolic lines perpendicular to  $L$ . Reflecting across both preserves all lines orthogonal to  $L$  and also fixes both endpoints. If you were to calculate the lengths, you would find that reflecting across two lines which are a distance of  $a$  apart (measured as the arc on  $L$ ), would translate by a distance of  $2a$ . Thus, any translation can be decomposed into reflections (circle inversions). If we let the endpoints be  $p_1, p_2$ , then we can draw the Euclidean triangle through  $p_1, p_2$  and  $x$ . There is a unique (Euclidean) circle passing through any three points, so let  $C'$  be the (Euclidean) circle passing through  $p_1, p_2, x$ . Since translation can be decomposed into reflections,  $C'$  must be fixed under any translation i.e.  $x$  will trace out the circle  $C'$ !

## 8. POINCARÉ DISC MODEL

Here, we discuss another model for hyperbolic space which illuminates different aspects of the theory. Instead of the upper half plane, this model is the points in the interior of a unit disc where the geodesics are semicircles orthogonal to the unit circle of the boundary (or diameters). To construct this model, we apply a circle invert to the upper half plane. We want to send  $\infty$  to the point  $(0, 1)$  and  $(0, 0)$  to  $(0, -1)$ . Thus, we need to invert across a circle centered at  $(0, 1)$  which sends  $(0, 0)$  to  $(0, -1)$ . Since the distance from  $(0, 1)$  to  $(0, 0)$  and  $(0, -1)$  is 1 and 2 respectively, we see that the radius of the appropriate circle is  $\sqrt{1 \times 2} = \sqrt{2}$ . If we invert through this circle, by general properties of circle inversion, we see that the  $x$  axis will be sent to a circle through  $(0, 1)$  and  $(0, -1)$  i.e. the unit circle centered at  $(0, 0)$ . Then, geodesics, which were circles perpendicular to the  $x$  axis, now become circles perpendicular to the unit circle. Note that while the Euclidean distance from any point to a point on the boundary is at most 2, the hyperbolic distance from any point to the unit circle is infinite. After all, this is just a model for lengths of hyperbolic space (or the pseudo-sphere if you would like).

One thing that this model allows us to see is which transformations leave a point fixed. If we rotate the unit circle around  $(0, 0)$ , it clearly preserves all geodesics and turns out to preserve lengths as well. From this, we can also see that points of hyperbolic distance  $d$  trace out a Euclidean circle around  $(0, 0)$  - pick one point  $p$  of hyperbolic distance  $d$  and rotate around  $(0, 0)$ , then the image of  $p$  will be another point of hyperbolic distance  $d$ . In fact, since any ray from  $(0, 0)$  has a unique point of distance  $d$ , we see that this circle is all the points which are a distance  $d$  away. From this, we can see that rotations and reflections through lines through  $(0, 0)$  are actually the only transformations that fix hyperbolic lengths and  $(0, 0)$ . They must both fix the origin and the circle corresponding to points a hyperbolic distance  $d$  away, which is just a Euclidean circle, it must be a rotation or a reflection through a diameter.

For any point  $p$  in the hyperbolic disc model, we can connect  $p$  to  $(0, 0)$  with a diameter of the unit circle. Then, we can translate along this line in much the same way we translated in the upper half plane - hyperbolic lines get sent to hyperbolic lines and points  $x$  trace out the (Euclidean) circle going through  $x$  and the two end points of the diameter. Thus, we can move  $(0, 0)$  to  $p$  by a translation which can be expressed as the composition of two hyperbolic reflections and hence preserves circles. Thus, we see that all hyperbolic circles are also Euclidean circles! Warning: the hyperbolic center of a hyperbolic circle does not coincide with the Euclidean center! The hyperbolic center will however be the intersection of all hyperbolic lines which are orthogonal to this Euclidean circle. By changing the hyperbolic radius of circles through  $p$ , we can foliate the disc model with hyperbolic circles centered at  $p$ . Similar to how we performed translations by translating on one line and moving the lines orthogonal to it, we can rotate one of these circles and rotate the lines orthogonal to it to get rotations around points in hyperbolic space. Thus, we see that all three basic transformations of Euclidean geometry - reflections, translations, and rotations all translate to Hyperbolic geometry very nicely.

Here's an interesting point: circles in Euclidean space which are not hyperbolic lines still represent something in hyperbolic space. If the circle is completely contained within the disc, it represents the points which are equidistant from *some point*. If the circle instead intersects the boundary twice, as we have seen, it is preserved under translations. Thus, it represents points which are equidistant with respect to *some line*.

Another very important thing that we can see from this picture is the angle between two hyperbolic lines coincides with the Euclidean angle. The reason for this is that the hyperbolic lines passing through  $(0, 0)$  are just diameters, and thus also Euclidean lines, so the hyperbolic and Euclidean angles match up. Now, given two hyperbolic lines  $L$  and  $L'$  intersecting at a point  $p$  in the disc model, we can translate  $p$  to  $(0, 0)$  and this translation preserves angles, both Euclidean and Hyperbolic. Euclidean angles are preserved because it's a composition of reflections, while hyperbolic angles are preserved because any symmetry of hyperbolic space preserves lengths and angles. Now, when it's translated to the origin, the angle between the two lines agree in the hyperbolic and Euclidean case. Thus, they must have agreed to begin with.

Here's another useful theorem: through any two hyperbolic lines which do not intersect and do not meet at the same point on the boundary of the disc model, there exists a unique hyperbolic line perpendicular to both. Say the two lines are  $L$  and  $L'$ . We can modify  $L$  by a sequence of symmetries so that it is the



vertical diameter in the disc model as follows: first take any point  $p$  on  $L$  and translate it to the origins so that  $L$  is a diameter. Then, rotate around the origin until  $L$  is the vertical line. Now, we show that with suitable translation along  $L$ , we can modify  $L'$  to be vertically symmetric, at which point we conclude that the horizontal diameter is our desired line. We may assume  $L'$  is on the right of  $L$  - both endpoints must be on the same side of  $L$  otherwise they would intersect, and if it's on the left, we can reflect through  $L$ . Label angles on the right half of the semicircle from  $-\frac{\pi}{2}$  at  $(0, -1)$  to  $\frac{\pi}{2}$  at  $(0, 1)$  with an angle of 0 at  $(1, 0)$ . Let the endpoints of  $L'$  have the angle values  $\theta_1, \theta_2$  and consider the sum  $\theta_1 + \theta_2$ . As we translate "upwards" along  $L$ , this sum will approach  $\pi$  as both points approach  $(0, 1)$  and as we translate "downward", the sum approaches  $-\pi$ . Thus, by the intermediate value theorem, there is some translation for which this sum is 0 i.e.  $\theta_1 = -\theta_2$ , or  $L'$  is vertically symmetric. Since symmetries do not change angles, we find that inverting all of our symmetries, the horizontal axis becomes orthogonal to the original  $L$  and  $L'$ , which we call  $\bar{L}$ .

To show  $\bar{L}$  is the unique line orthogonal to  $L$  and  $L'$ , note that reflecting along  $L$  and  $L'$  is a translation along  $\bar{L}$ . If there were another line that were orthogonal to both  $L$  and  $L'$ , say  $\bar{L}'$ , then the reflection would have to be simultaneous translation along both lines. Doing these reflections repeatedly, we approach one of the end points of  $\bar{L}$  and thus also  $\bar{L}'$ , while doing the reflections in the other order, we approach the other endpoint of both lines. Thus, both lines share both endpoints on the unit circle, and thus must be the same line. Note: this line allows us to define the distance between lines  $L$  and  $L'$  by taking the length of the arc between them on the line perpendicular to them both. It is an exercise to show that this is in fact the minimum length between any pair of points  $p \in L$  and  $p' \in L'$ .

Note that since the lines don't intersect, they're parallel. In Euclidean space, we have that the reflection through two parallel lines is a translation through a line perpendicular to both. Here, even though there's infinitely many lines parallel to any one line, the same statement holds! Thus, calling two lines which don't intersect parallel has actual geometric significance! However, there is a distinction: in Euclidean space, there are infinitely many line orthogonal to two lines, while we have just shown that in Hyperbolic space, there is just one line orthogonal to two lines. This is somewhat the opposite of how in Euclidean space, there is one line parallel through another with a given point, while in Hyperbolic space, there's infinitely many.

[Reflection of Non-Parallel Lines] What happens if you reflect across two lines which intersect in the disc? Is this analogous to how things work in Euclidean space? What happens if two points do not intersect in the disc, but both approach the same endpoint on the unit circle (necessarily, the other endpoint is different)?

I should point out that there are most certainly not the only ways to prove these statements. If they don't speak to you, try playing around with the hyperbolic plane and coming up with your own proofs!

## 9. BACK TO TOPOLOGY!

We can now answer the question of if we can accurately chart a surface of genus  $g \geq 2$ . By the Gauss-Bonnet theorem, we know that such a parametrization must have negative curvature. So the question arises: can we parametrize the surface using the hyperbolic plane, which has constant negative curvature? The answer is yes!

We will sketch this for a genus 2 surface by showing that you can tile the hyperbolic plane with octagons. Here's the argument: since the 8 vertices of the octagon are all identified at the same point, their angles must sum up to  $2\pi$ , and so each one must have an angle of  $\frac{\pi}{4}$  (if it is a regular octagon). Now, take the Poincare disc model and draw 8 circles of the same radius which are orthogonal to the unit circle at the boundary and all tangent to one another. This forms an "ideal octagon" whose vertices are actually points infinitely far away (called "ideal points") and the angles between adjacent hyperbolic lines is 0, since they are tangent. Now, increase the radius of these circles while fixing their Euclidean center until they are almost 1. Then, they will trace out an octagon very close to the origin. The hyperbolic segments tracing out the octagon will be almost Euclidean straight, and thus each angle will be very close to the Euclidean angle of a regular octagon, that is  $\pi \frac{6}{8} = \frac{3\pi}{4}$ .

As we increased the radius of the circles, the interior angles went from 0 to  $\frac{3\pi}{4}$ , so by the intermediate value theorem, there must have been some radius where the interior angles of the regular octagon were  $\frac{\pi}{4}$ ! This is a candidate octagon which is the fundamental region of some tiling of the hyperbolic plane. How do we construct this tiling? Notice that we need to identify every other edge with "opposite orientation".

This can be achieved as follows: suppose we want to identify the lines  $L$  and  $L'$  (say both correspond to the label  $a$ ), we can take the line  $Q$  that is perpendicular to both and translate along it until  $L'$  coincides with  $L$ . Translating the whole octagon adds a new octagon to the hyperbolic plane where the  $a$  of  $L$  gets identified with the  $a$  of  $L'$  in the correct orientation. We can now do this with a different pair of sides of our original octagon, or with a pair of sides of our new octagon. Repeating this process indefinitely will tile our plane with these octagons such that the one we constructed is a fundamental domain. Quotienting hyperbolic space by this collection of transformations generated by translations then gives us a genus 2 surface with constant  $-1$  curvature! It's left as an exercise that this can be done for any surface of genus  $g \geq 2$ . What happens if you take an ideal square? An ideal triangle?